

SIDDIGAN TM

Data Engineer

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Driven Data Engineer with experience in fast-paced Agile development environment to design, build, deploy and manage data pipelines, ETL processes, and data warehousing solutions. Skilled in Python, SQL, Snowflake and AWS, with working knowledge of other data tools. Eager to apply my technical skills to tackle realworld data challenges and contribute to Data, cloud and AI projects.

PROFESSIONAL EXPERIENCE

Data Engineer, ZS 2025 – present
Gurgoan

- **Designed and implemented various layers of a comprehensive Pharma Data Lake**, from scratch by driving the process from requirement gathering and data profiling to data modeling with Erwin, tailored to regulatory data use cases.
- **Designed and deployed scalable, end-to-end ETL pipelines** using Apache Airflow, AWS EMR, AWS Glue, PySpark, and SparkSQL across the data lake, ensuring efficient ingestion, transformation, and supporting advanced analytics and reporting.
- **Built robust ETL testing and reconciliation frameworks**, including functional, regression, and data quality validations, ensuring accuracy, consistency, and compliance with pharma industry standards.
- **Enhanced and optimized existing data architecture by applying changes aligned with evolving pharmaceutical business needs** and proactively addressing potential future use cases such as clinical trial data integration and real-world evidence analytics

Data Engineer - Apprentice, Fidelity Investments 2024 – 2025
Bengaluru, India

- **Played a key role in the cloud migration of enterprise data lake to the cloud**, designing and optimizing end-to-end data pipelines for the Data Lake. Transitioned data Oracle DB systems into Snowflake with AWS.
- **Developed and implemented curated tables, stored procedures, and dynamic scripts**, significantly enhancing ETL processes, ensuring robust data validation, and automating data flows. Leveraged Control-M, AWS Lamda and Apache NiFi to streamline workflows and scale data processing across enterprise systems.
- **Created and maintained operational Power BI dashboards**, improved operational efficiency and delivered business impact that reduced failure rate by 78%.
- **Worked on a POC for data masking Snowflake DB using Tonic AI**, which included designing, building, testing, and conducting thorough risk and impact analyses to support data privacy and compliance initiatives for data governance.
- **Conducted regression testing and wrote complex queries for ICEDQ** to validate tables and views, ensuring data quality and accuracy across critical data processes.

SKILLS

Big Data Engineering | Data Warehousing | ETL / ELT | Snowflake | AWS (EMR, Glue, Athena, S3) | Apache Airflow | Apache NiFi | Data Privacy / Masking | ICEDQ | Git | Pyspark / SparkSQL | Data Pipeline Design | Data Analysis | PostgreSQL / SnowSQL | Python

PROJECTS

Financial Advisory and Consultancy Chatbot 2024

- **Developed a financial advisory chatbot** using a fine-tuned **Llama 3 model** integrated with a **Retrieval-Augmented Generation (RAG) system**, fetching and analysing real-time stock metrics and market news for personalized investment advice.
- **Engineered data pipelines** using **Apache NiFi** and **Airflow** to automate daily updates to **PostgreSQL** and **vector DB**, optimizing the bot's ability to retrieve and analyze financial data for intelligent, context-driven responses.

Resume Matching System

- **Built a resume matching system** using **BERT embeddings** to accurately rank and identify top candidate matches from resumes and job descriptions, optimizing recruitment and job search processes.

2023

EDUCATION

Bachelor of Technology: Computer Science Engineering , <i>Lovely Professional University</i> Machine Learning Specialization	2019 – 2023 Phagwara
High School Intermediate , <i>Narayana Junior College</i> Secured 83.5%	2017 – 2019 Hyderabad

PUBLICATIONS

TDISE (Text-Driven Image Synthesis and Editing) , <i>SSRN</i> Proposed the TDISE model, which synthesizes and edits images based on descriptive text. The model combines elements from SINE, TediGAN, and GLIDE, CLIP encoder, StyleGAN, and advanced inversion and optimization techniques for precise text-guided image transformations, contributing to advancements in image synthesis research.	2024
Pixelforge: Image Revamping Through Text and Image Input Using Pix2Pix , <i>Taylor & Francis</i> Pixelforge over see significant steps forward in image editing and generation. showcasing a fine-tuned InstructPix2Pix models that enables intuitive, mask-free image editing through text prompts, making it accessible for non-experts and customizable for diverse applications as a review paper.	2024

TRAINING AND CERTIFICATIONS

• Amazon Machine Learning Summer School '22	• TensorFlow Developer Professional Certificate - DL.AI
• Crash Course on Python - Google	• IBM Qiskit Global Summer School '23

INTERESTS

Data Engineering | Data Science | Generative AI | Quantum Computing | Particle Physics | Photography